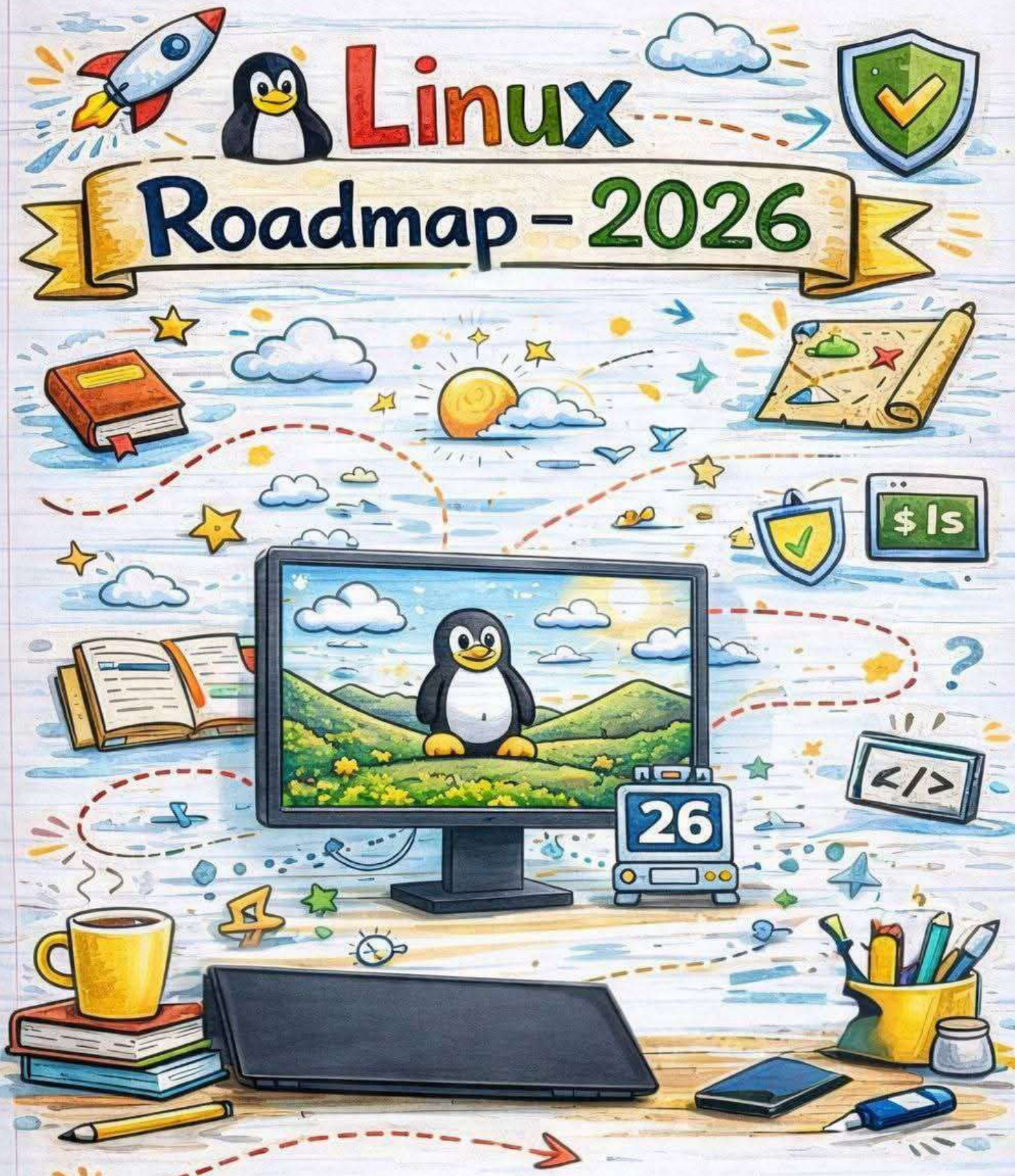
Linux Roadmap - 2026



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Introduction to Linux

What is Linux?

Linux is:

- An Open-Source Operating System

Used in:

- ✅ Servers
- ✅ Cloud Platforms
- ✅ Supercomputers
- ✅ Embedded Systems

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Why Learn Linux?

- ✅ Most servers run on Linux
- ✅ Essential for DevOps & Cloud
- ✅ Free & Secure
- ✅ High Demand Skill

Who Should Learn Linux?

- Students
- Developers
- Sys Admins

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Linux Installation & Distributions

Linux Distributions

Popular distros:

- ✔ Ubuntu
- ✔ Debian
- ✔ CentOS
- ✔ Fedora

- Choose at boot

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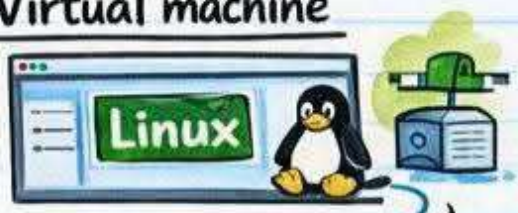
Installation Methods

✔ Dual boot

- Choose at boot

✔ Virtual machine



- Runs inside host OS

Linux Interface

✔ GUI (Graphical)

- Point & Click

CLI (Terminal)

- Type Commands

Important Linux Directories

- /home — User files & configs
- /etc
- /etc System config files
- /var Logs, temp data
- /bin Important commands

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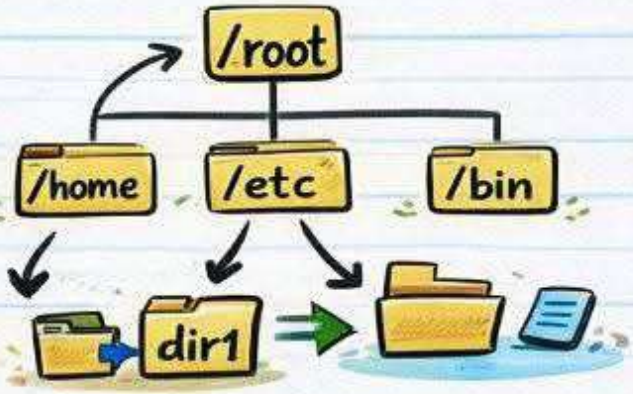
Linux File System & Commands

File System Basics

Linux uses:

- Hierarchical file system



Basic Commands:

- **ls** - list files
- **cd** - change directory
- **pwd** - print working directory → /home/user1/dir1
- **mkdir** - make new directory

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File Operations

- **touch** - create a new file → detailed view
- **cp** - copy a file → move or rename a file
- **mv** - move or rename a file → $ cd di/ /dir1

Command Tips

Use flags ls -L
- detailed view

Use tab completion
- auto-completes commands

cd cd di /dir1

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Users, Groups & Permissions

Users & Groups

- Linux is:
  - Multi-user operating system

Permissions

Three types:

- Read (r) - view a file
- Write (w) - modify a file
- Execute (x) - run a file/program

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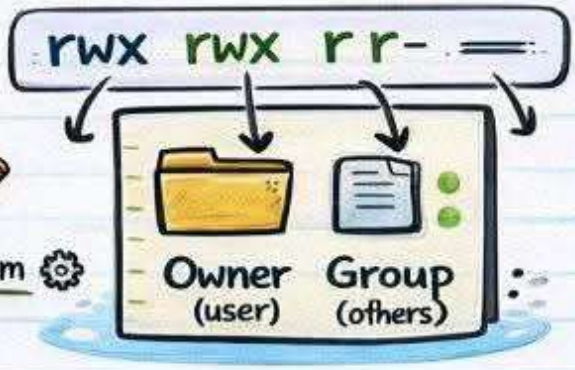

Permission Groups

- Owner (user)
- Group

Permission Commands

- chmod 777 file1
  - change permissions
- chown ...
  - change owner

Why Permissions Important?

- Security & access control

```
$ chmod 777 file1
```

```
$ chown user1 file1
```

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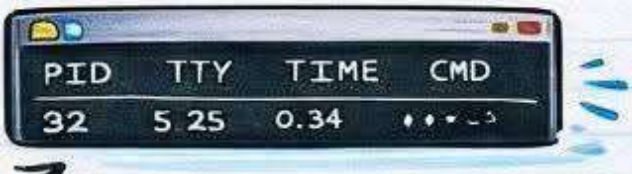
Processes, System Monitoring & Networking

Processes

- Process:
- A running program

Process Commands

ps – list running processes



top - real-time process viewer

htop (advanced)

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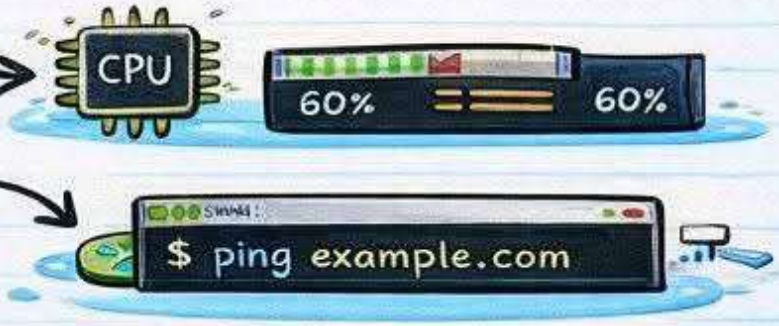
System Monitoring

- ☑ CPU usage
- ☑ Memory usage
- ☑ Disk usage



Networking Basics

✔ IP address

192.168.1.1-1

Ping

Ports

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Package Management & Shell Scripting

Package Management

Used to:

- ✅ Install software
- ✅ Update software

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Package Tools

apt → `$ apt install`

yum → `$ yum install`

dnf → `$ dnf install`

Shell Scripting

✅ Shell script:

- Automates tasks

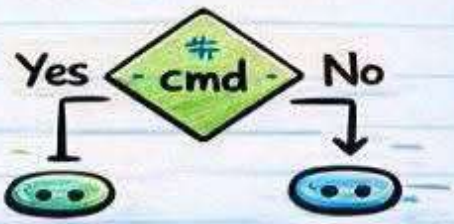
Script Basics

✅ Variables

```
$ name=username
$ count=5
$ flag=true
```

✅ Loops

✅ Conditions



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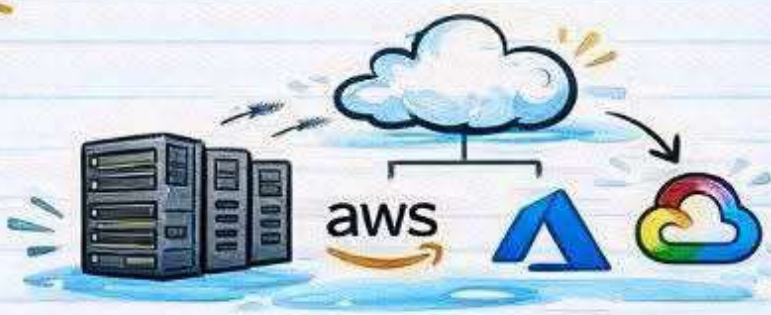
Linux for DevOps, Cloud & Security

Linux in Cloud

- Used in:
  - ✅ AWS
  - ✅ Azure
  - ✅ GCP



Linux Security

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- ✅ Firewall basics
- ✅ Secure access
- ✅ Updates

Linux in DevOps

- ✅ Servers
- ✅ Containers
- ✅ Automation

Real-World Usage

- ✅ Hosting websites
- ✅ Running APIs

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Linux for DevOps, Cloud & Security

Roadmap, Practice & Career Path

Linux Learning Roadmap

Basics → Commands → Permissions → Processes →

Practice Ideas

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- ☑ Create users
- ☑ Write scripts
- ☑ Manage files

Career Roles

- ☑ Linux Administrator
- ☑ Cloud Engineer
- ☑ System Engineer

Final Tips

- ☑ Practice in terminal daily
- ☑ Learn commands deeply
- ☑ Focus on security
- ☑ Stay consistent
- ☑ Automate tasks

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